

IMECE2013-63711

A REDUCED-ORDER MODEL OF A MICROFLUIDIC TRANSISTOR

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ABSTRACT

A reduced-order model of a Microfluidic Transistor is presented. The transistor is essentially a long micro channel between substrate and a membrane that is pressure actuated. The proposed model captures steady (DC) and small signal (AC) behavior of the device in a manner analogous to standard semiconductor transistor models. The model is based on steady and perturbed unsteady solutions of the conservation of mass and momentum, coupled with an elastic model for the membrane. To improve the accuracy and to enhance the range of validity, the model is enhanced by numerical simulations of the coupled fluid-structure problem. The model predicts dependence of the transconductance on the pressure differentials across the membrane and along the channel. The proposed model also investigates the impact of flow inertia, among other effects, on the dynamic behavior of the transistor.

NOMENCLATURE

$\dot{m}$ Mass flow rate
 t time
 μ Viscosity of the fluid

E Modulus of elasticity
 L Length of fluidic transistor
 L_c Length of channel before fluidic transistor
 h_0 Height of undeflected fluidic transistor
 w Width of fluidic transistor
 ρ Density of the fluid
 U Velocity field inside the fluidic transistor
 d Thickness of the membrane
 δ Vertical deflection of the membrane from equilibrium
 δ_c Vertical deflection at the center of the membrane
 u Velocity of fluid in the x direction
 v Velocity of fluid in the y direction
 Q Volume flow rate
 g_m Transconductance
 r_o Intrinsic output resistance
 A_v Intrinsic pressure gain
 p_{in} Inlet pressure of the fluid
 p_{out} Outlet pressure of the fluid
 p_{ext} External pressure above the membrane
 $\bar{p}$ Average pressure inside fluidic transistor
 δp Pressure drop from input to output
 Δp Actuation Pressure relative to output pressure

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INTRODUCTION

Extensive research has been done on characterizing microfluidic components and circuits in the last few years, due to the wide spectrum of applications of microfluidic devices, especially in biotechnology. One component in such circuits that is gaining increasing attention is the elastic membrane micro channel. Typically, elastic membranes have been considered as capacitors because they can store mass [1, 2]. On the other hand, the same component can be used as a switch when enough pressure on the membrane blocks the flow. In addition, those devices are used as micro pumps for biological systems that simulate or replace biological transport phenomena [2]. As [3] states, there is a need for simple means to translate an electric signal to a fluidic signal without the use of electromechanical interface actuators. The existence of a large number of actuators is a problem especially when designing large integrated microfluidic circuits. A solution was suggested by Leslie [1] who managed to control the output flow rate by pressure actuating fluidic capacitors in parallel branched to channels that have different resonance frequencies. This strategy diminishes the need for a large number of actuators and shifts the control to frequency actuation. A similar device has also been used as a switch to model microfluidic logic circuits [4–7]. In these papers, the authors have introduced transistor-like elements as the basic building blocks for their logic circuits. In [5] the transistors were pneumatically driven, while in [6] they were fluid driven. Analogies between electric components and fluidic elements have been extensively used in fluidic circuit analysis [8].

Two different analysis procedures were proposed by [9, 10]. Utz [9] designed a piezoelectric actuated diaphragm micro pump and derived a linearized model for its dynamic response and performance. [10] focuses on characterizing the propagation of pressure waves actuated by elastic membranes to control the flow in micro channels. Utz [9] characterizes mathematically the frequency response of a long channel covered by an elastic membrane. The model investigates the pressure waves propagation along a long passive membrane assuming a parabolic profile deflection transverse to the flow. These two papers focus on the capacitive behavior of such channels. Although these components can be found frequently in the literature [1, 4–6, 11] they haven't been thoroughly characterized mathematically and experimentally.

In this paper, we propose a reduced order analog model for the microfluidic transistor depicted in Fig. 1. The transistor consists of a microchannel where the fluid flowrate is controlled by the combined effect of the pressure differential across the channel ($p_{in} - p_{out}$) and the upper wall deflection (δ) controlled by the external pressure (p_{ext}). A reduced-order model is an analytical model that captures the essential physical behavior of a device

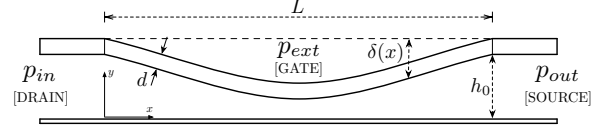


FIGURE 1. SCHEMATIC OF MICROFLUIDIC TRANSISTOR.

while employing some assumptions to reduce the complexity of the analysis [12]. A reduced-order model is efficient and accurate just enough for preliminary design purposes offering advantages in terms of speed, cost and ability to perform analysis and design in an integrated multiple domain “circuit design” environment such as Spice [13]. In addition it is much more practical and workable than tables and numerical models [12]. The parameters of the resulting reduced order model can be determined analytically, usually under some simplifying approximations, or more accurately from detailed numerical simulation of the problem. In this work, we first present a simplified analytical model for the device. Second, we employ this derived analytical model to build a reduced order analog model that captures the essential behavior of the device. Third, we present a more accurate reduced order model parameters determined from detailed numerical simulations using FSI (Fluid Structure Interaction) simulations on FLUENT/ANSYS. Finally, we compare the performance of the reduced order model based on the simplified analytical model with that with parameters obtained numerically.

ANALYTICAL MODEL

We first present a model for the membrane deflection which is then used in deriving the equation governing dependence of the volume flowrate on various parameters.

Membrane model

In order to model the fluid dynamics of the problem, we need to use a deflection model of the membrane. Various membrane deflection models can be found in the literature. Senturia [12] assumes a cosine trial function and uses variational methods to derive the deflection model of a square membrane under uniform loading. Although Senturia's model is for a 2D plate, we are interested in the behavior of a section of the plate while taking into account 3D effects (induced by the poisson ratio ν). Beggley [14] derives a linear and a cubic model for small and large deflections of a PDMS membrane. Both models account for the linear dependence of deflection on loading at small deflections due to bending and a cubic dependence for large deflections due to stretching. The main difference is that [14] splits the model into polynomials while Senturia includes both aspects in the same equation. Fig. 2 shows the deflection along the length

of the membrane for different loading conditions ($M = \frac{pL^3}{Ed^3}$, $L = 3118\mu\text{m}$, $d = 159\mu\text{m}$, $E = 26.7\text{ MPa}$, $p = \bar{p} - p_{ext}$) as predicted by [12] and the small and large deflection models in [14], where $\bar{p}$ is the average pressure in the channel. While the two

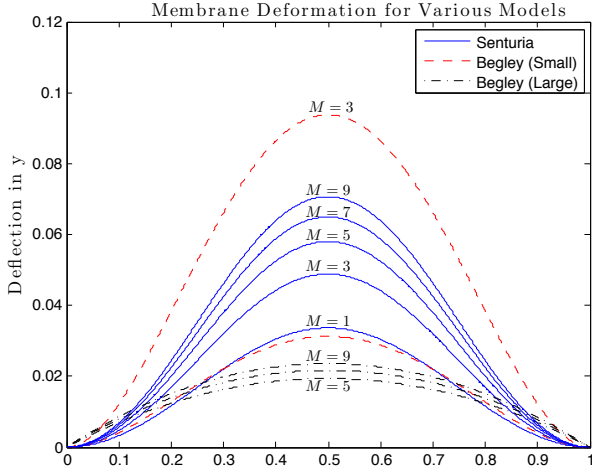


FIGURE 2. MEMBRANE DEFLECTION FOR VARIOUS ANALYTICAL MODELS

models agree for small deflections, they differ at large deflections. The model presented in [12] assumes a deflection of the form

$$\delta(x, t) = \frac{\delta_c(t)}{2} \left(1 - \cos\left(\frac{2\pi x}{L}\right) \right) \quad (1)$$

that automatically satisfies the boundary conditions at both fixed ends. This model [12] accounts for both small and large deflections according to the following relation

$$p^*(t) = \bar{p} - p_{ext}(t) = \frac{2\pi^4}{3} \left(\frac{Ed^3}{(1-\nu^2)L^4} \right) \delta_c(t) + \frac{\pi^4}{4} \frac{(7-2\nu)(5+4\nu)}{32(1+\nu)} \left(\frac{Ed}{(1-\nu)L^4} \right) \delta_c^3(t) \quad (2)$$

where δ_c is the deflection of the center of the membrane.

Fig. 3 shows a comparison of the maximum deflection of the center of the membrane as predicted by Eqn. (2) to that predicted by numerical simulation where the pressure distribution acting on the fluid side of the membrane is due to steady flow of the liquid under the membrane. The departure of the model (Eqns. (1) - (2)) from numerical simulation is due to the assumptions underlying the analytical model. The model assumes small symmetric

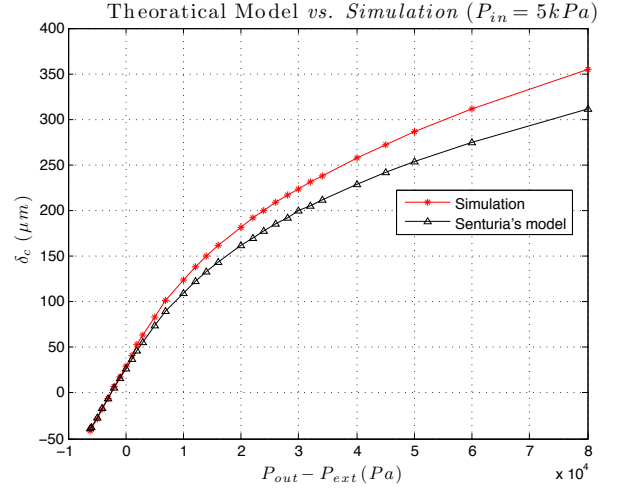


FIGURE 3. VALIDITY OF THEORETICAL MODEL SUBJECT TO SIMILAR CONDITIONS AS SIMULATION

deflection of the membrane with uniform pressure distribution on both sides of the membrane. These conditions differ from the actual case where the pressure on the fluid side varies along the flow direction from the “drain” to the “source”. The difference between the flow pressure and that on the “gate” side may be large enough so that the small deflection assumption is violated. Therefore, the difference between the proposed reduced order model and the numerical simulation becomes more prominent for larger values of Δp , δp and δ_c . Given the simplifications underlying the model, the agreement between the model and simulation in terms of the center membrane deflection shown in Fig. 3 is acceptable.

Fluid channel model

We consider incompressible, inertia-free flow in a 2D channel of length L and thickness $h = h_0 + \delta$. For $h \ll L$, the momentum equation in the x -direction may be approximated as [15]

$$0 \simeq -\frac{\partial p}{\partial x} + \mu \frac{\partial^2 u}{\partial y^2} \quad (3)$$

For $\delta \ll h_0$, v scales as $\partial h / \partial t$. If the pressure gradient is such that $u \gg v$, then we may assume from the momentum equation in the y -direction, that p is approximately independent of y . For $p(x, t)$, we may then integrate Eqn. (3), subject to no-slip boundary condition at $y = 0, h$, to get

$$u = \frac{1}{2\mu} \left(-\frac{\partial p}{\partial x} \right) (h^2 - y^2) \quad (4)$$

The volume flowrate is then

$$Q(t) = w \int_0^h u dy = \frac{wh^3}{12\mu} \left(-\frac{\partial p}{\partial x} \right) \quad (5)$$

Noting that $h(x,t) = h_0 + \delta(x,t)$ then $h^3 \simeq h_0^3(1 + 3\delta/h_0)$ for $\delta \ll h_0$, and using Eqn. (1)

$$Q(t) \simeq \frac{wh_0^3}{12\mu} \left(-\frac{\partial p}{\partial x} \right) \left(1 + \frac{3\delta_c(t)}{2h_0} \left(1 - \cos\left(\frac{2\pi x}{L}\right) \right) \right) \quad (6)$$

For δ_c very small, Eqn. (2) may be approximated as $p^* \simeq C \delta_c(t)$ where $C = \frac{2\pi^4}{3} \left(\frac{Ed^3}{(1-\nu^2)L^4} \right)$. Therefore

$$Q(t) \simeq \frac{wh_0^3}{12\mu} \left(-\frac{\partial p}{\partial x} \right) \left(1 + \frac{3p^*(t)}{2h_0C} \left(1 - \cos\left(\frac{2\pi x}{L}\right) \right) \right) \quad (7)$$

and

$$-\frac{\partial p}{\partial x} = \frac{12\mu Q(t)}{wh_0^3} \left(1 + \frac{3p^*(t)}{2h_0C} \left(1 - \cos\left(\frac{2\pi x}{L}\right) \right) \right)^{-1} \quad (8)$$

Integrating from $x = 0$ (at which $p = p_{in}$) to $x = L$ (at which $p = p_{out}$), we arrive at the pressure drop in the channel

$$p_{in} - p_{out} = \frac{12\mu Q(t)}{wh_0^3} \frac{L}{\sqrt{1 + \frac{3p^*(t)}{h_0C}}} \quad (9)$$

Then

$$Q = \frac{wh_0^3}{12\mu L} \delta p \sqrt{1 + \frac{3p^*(t)}{h_0C}} \quad (10)$$

where $\delta p = p_{in} - p_{out}$. Approximating $p^* = \bar{p} - p_{ext} \simeq (p_{in} + p_{out})/2 - p_{ext} = \delta p/2 + \Delta p$, where $\Delta p = p_{out} - p_{ext}$, we arrive at the equation governing the fluid volume flowrate

$$Q = \frac{wh_0^3}{12\mu L} \delta p \sqrt{1 + \frac{3}{h_0C} \left(\frac{\delta p}{2} + \Delta p \right)} \quad (11)$$

REDUCED ORDER ANALOG MODEL OF THE MICROFLUIDIC TRANSISTOR

It can be observed from Eqn. (11) that, at low frequencies (in absence of inertia effects), the device operates like a transistor with a reduced order model for small signal operation as

shown in Fig. 4. This model corresponds to a pressure (voltage) controlled flowrate (current) source, with an intrinsic output resistance r_o . By small signal operation, we mean transient operation of the device in the neighborhood of a steady operating point (DC), under which conditions the pressure differential and flowrate may be expressed as

$$\Delta p(t) = \Delta p_{DC} + \Delta' p(t) \quad (12)$$

$$\delta p(t) = \delta p_{DC} + \delta' p(t) \quad (13)$$

$$Q(t) = Q_{DC} + Q'(t) \quad (14)$$

where $|\Delta' p| \ll \Delta p_{DC}$, $|\delta' p| \ll \delta p_{DC}$, and $|Q'| \ll Q_{DC}$. It should be mentioned that although the electro-hydraulic analogy is a very powerful tool for reduced order modeling, it is important to recognize the limitations of this approach due to the fundamental differences between mechanical and electrical systems [16]. Since the behavior is similar to an electronic MOSFET, we can go further into defining small signal parameters for our fluidic transistor such as the transconductance, intrinsic output resistance, and intrinsic gain by making the following analogies with the electronic transistor

$$g_m = \frac{\partial I_D}{\partial V_{GS}} \leftrightarrow \frac{\partial Q}{\partial (\Delta p)} \quad (15)$$

$$r_o = \left(\frac{\partial I_D}{\partial V_{DS}} \right)^{-1} \leftrightarrow \left(\frac{\partial Q}{\partial (\delta p)} \right)^{-1} \quad (16)$$

$$A_v = g_m r_o \quad (17)$$

As mentioned before, the reduced order model parameters g_m

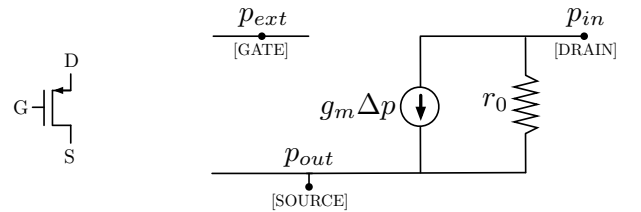


FIGURE 4. LOW FREQUENCY SMALL SIGNAL MICROTRAN-SISTOR ANALOG MODEL

and r_o can be obtained analytically from the simplified model of Eqn. (11) or more accurately determined from numerical simulations of the fluid-structure interaction, as presented in the following section.

NUMERICAL RESULTS

In this section, we present the steady behavior of the 2D pressure-actuated microfluidic transistor depicted in Fig. 1, using FSI (Fluid Structure Interaction) simulations on FLUENT/ANSYS over ranges of operating pressures ($\delta p = p_{in} - p_{out}$ and $\Delta p = p_{out} - p_{ext}$) to construct accurate characteristic curves analog to those of drain current versus drain-source and gate-source voltage differences in a MOSFET. These characteristic curves are then used as the basis for choosing the “DC” operating point of the transistor, and extraction of small signal parameters such as transconductance, intrinsic output resistance, and pressure gain. The large signal steady (DC) behavior as well as the small signal unsteady (AC) behavior will be compared and contrasted with those of a standard MOSFET. The numerical results are obtained from ANSYS/FLUENT numerical simulations of the coupled fluid-structure interaction. At each time step, the FLUENT solution of the pressure distribution in the fluid is used by ANSYS to solve for the deflection of the membrane, which is then used to update the mesh in FLUENT for the following time step, and so on. Although the simulation is performed on a 3D setup, we impose symmetry in the depth direction making it only $(1 - \nu^2)$ times different than any of its 2D sections. The mesh is made of cubic elements having a maximum size of $10\mu\text{m}$. The mesh size is chosen so that the solution is nearly mesh-independent. In addition, the mesh is dynamic, changing in size as the solid and the fluid domains deform, in order to account for the high gradients of pressure, especially when the membrane is squeezed down to the floor of the device. The membrane is fixed from both sides and p_{in} , p_{out} and p_{ext} are fixed for each simulation. In our simulations, the length of the membrane is $L = 3118\mu\text{m}$ with a channel height of $h_0 = 59\mu\text{m}$ and a membrane thickness of $d = 159\mu\text{m}$ similar to those used in [1]. The model is simulated in 3D with a channel width of $w = 500\mu\text{m}$ with careful control on 3D effects by imposing symmetry.

Operating Point (DC) Characteristic Curves

From a phenomenological point of view, we expect the rate of increase of mass flow rate to decrease as the deflection increases, because more work is required to stretch it. This non linear behavior, shown in Fig. 5, is predicted in the numerical simulations as we increase p_{ext} for a fixed p_{in} . It is also important to note that for small p_{in} , $\frac{\partial Q}{\partial \delta p}$ seems to stay constant, showing that the stretching effect is still small. Therefore, there is a threshold of p_{in} after which the stretching term becomes significant and the membrane cannot be assumed to be in pure bending. Following the procedure conventionally employed in small signal characterization of electronic transistors [17], we start by plotting the volume flow rate Q versus $\delta p = p_{in} - p_{out}$ for fixed values of $\Delta p = p_{out} - p_{ext}$, in analogy to plotting I_D versus V_{DS} for fixed values of V_{GS} . Note that we have $\Delta p = p_{out} - p_{ext}$ analogous to V_{GS} in the graphs because of the sign convention of p_{ext} . As in

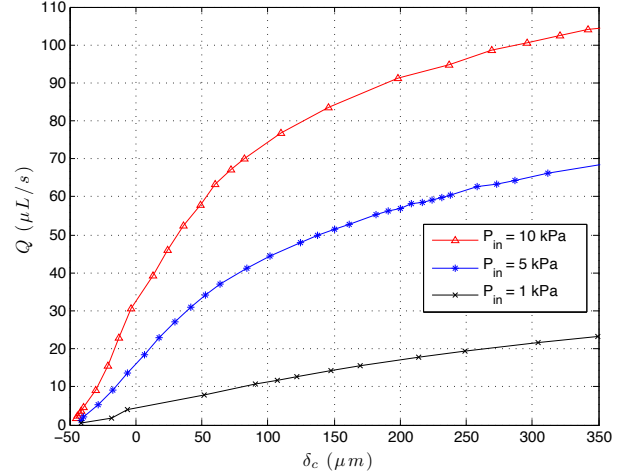


FIGURE 5. EFFECT OF DEFLECTION ON VOLUME FLOW RATE

a MOSFET, the expected trend of the curve is to level down for large values of Δp with a parabolic triode region. The microfluidic transistor exhibits a similar behavior, as can be observed in Fig. 6 for small δp and $p_{ext} > 0$ (for example $0 \leq \delta p \leq 30 \text{ kPa}$ and $p_{ext} = 10 \text{ kPa}$). Note that for $p_{ext} < 0$, the membrane is deflected upwards, and the device behaves as a capacitor. It can be

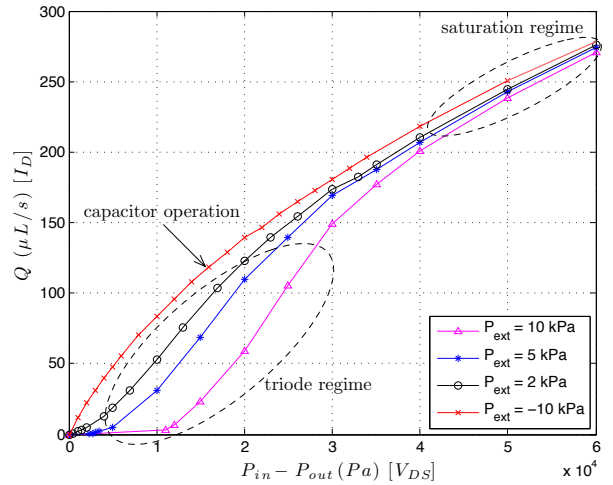


FIGURE 6. $Q - \delta p$ CHARACTERISTICS OF THE MICROFLUIDIC TRANSISTOR

further noticed that, for $\delta p > \delta p_{th}$ where δp_{th} is some threshold pressure differential across the channel that is a function of Δp , the flow rate exhibits linear dependence on the pressure dif-

ferential across the channel (δp) as it becomes dominant with a decreasing dependence on p_{ext} . This linear behavior is similar to the electronic transistor behavior in the saturation regime where the drain-source voltage exceeds some threshold value.

Typically, MOSFETs have limitations on the operating voltage because of the so-called short channel effect which arises when the channel length is of the same order of magnitude as the depletion-layer width of the source and drain junction. One of the manifestations of such an effect is the saturation of the velocity of electrons. Interestingly, the short channel effects shows up in our microfluidic transistor whereby the mass flow rate saturates at some point. In order to identify the different operating modes of a MOSFET, I_D is plotted versus V_{GS} for fixed values of V_{DS} . Analogously, we plot Q versus $p_{out} - p_{ext}$ for fixed values of $p_{in} - p_{out}$ as shown in Fig. 7. And it can be noticed that the trend is parabolic in the first region, i.e. long channel region, and linear in the second region, i.e. short channel region, as in the MOSFET.

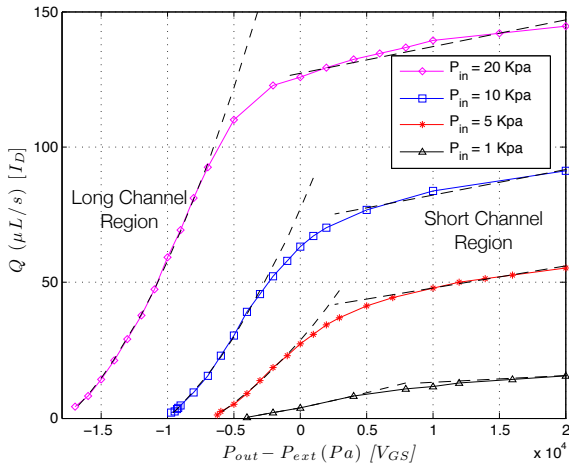


FIGURE 7. $Q - \Delta p$ CHARACTERISTICS OF A FLUIDIC TRANSISTOR

Small Signal Parameters

The numerical results for the dependence of the volume flow rate on δp and Δp , plotted in Figs. 6 and 7 respectively, are used in conjunction with Eqns. (15)-(16) to compute the small signal parameters of the reduced order model. It can be seen in Fig. 8 that, for a given δp , the fluidic transconductance attains a maximum for some characteristic value of Δp . It may be of interest to operate at the point of maximum transconductance to maximize the small signal component of the flowrate for an applied pressure difference. The intrinsic output resistance, computed using

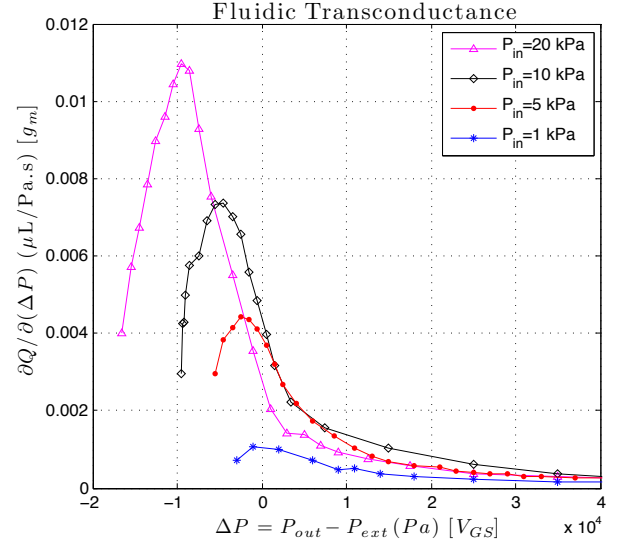


FIGURE 8. FLUIDIC TRANSCONDUCTANCE

Eqns. (16), is plotted in Fig. 9 versus δp for different values of Δp . It can be observed the device exhibits large intrinsic resistance (to small signal variation in δp) for small values of δp and $p_{ext} > 0$ with this resistance increasing for larger external pressure. This resistance drops rapidly as the pressure differential across the channel is increased and obeys linear dependence on δp beyond a threshold value.

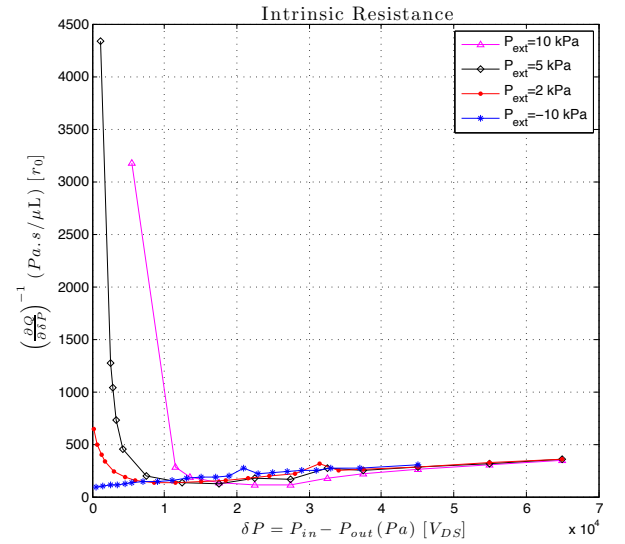


FIGURE 9. INTRINSIC RESISTANCE OF MICROFLUIDIC TRANSISTOR

Comparing Analytical Model to Numerical Results

The reduced order model parameters computed from the numerical simulations in the previous section are compared with those predicted by the simplified analytical model that is based on inertia-free flow for small deflection of the membrane. Although the analytical model predicts the qualitative behavior of the transistor for small deflections, and more accurate characterization of the microfluidic transistor is obtained by carrying out numerical simulations, the analytical model provides insight into the physics underlying the behavior of the device over the range of parameters for which the analytical model is accurate.

The reduced order model parameters, i.e. the fluidic transconductance and intrinsic output resistance, predicted by the analytical model (Eqn. (11)) are

$$g_m = \frac{\partial Q}{\partial(\Delta p)} = \frac{wh_0^2}{8\mu LC} \frac{\delta p}{\sqrt{1 + \frac{3}{h_0 C} \left(\frac{\delta p}{2} + \Delta p \right)}} \quad (18)$$

$$r_o = \left(\frac{\partial Q}{\partial(\delta p)} \right)^{-1} = \frac{12\mu L}{wh_0^3} \sqrt{1 + \frac{3}{h_0 C} \left(\frac{\delta p}{2} + \Delta p \right)} \quad (19)$$

The intrinsic pressure gain (analogous to small signal voltage gain) is

$$A_v = g_m r_o = \frac{3}{2h_0 C} \frac{\delta p}{1 + \frac{3}{h_0 C} \left(\frac{3\delta p}{4} + \Delta p \right)} \quad (20)$$

The transconductance and intrinsic resistance predicted by the analytical model are shown along those computed from numerical simulations in Fig.10 and Fig.11 respectively, for a small range of Δp and δp . It can be observed that the analytical model does not predict the maximum in the transconductance, otherwise both models agree qualitatively. The analytical model assumes small deflections and small perturbations of the membrane and a uniform pressure distribution under the membrane. Therefore, the analytical model presumes a nearly parallel flow (undergoing a nearly Poiseuille flow), which explains the divergence of the theoretical and experimental model for large Δp and large δp .

Small Signal Transient Operation

In this section, we investigate small signal transient behavior of the common-source configuration shown in Fig. 12, where a channel (of $750\mu\text{m}$ length) is connected in series with the drain of the transistor. A small harmonic sinusoidal perturbation is applied to the membrane. The total gain of the circuit is the product

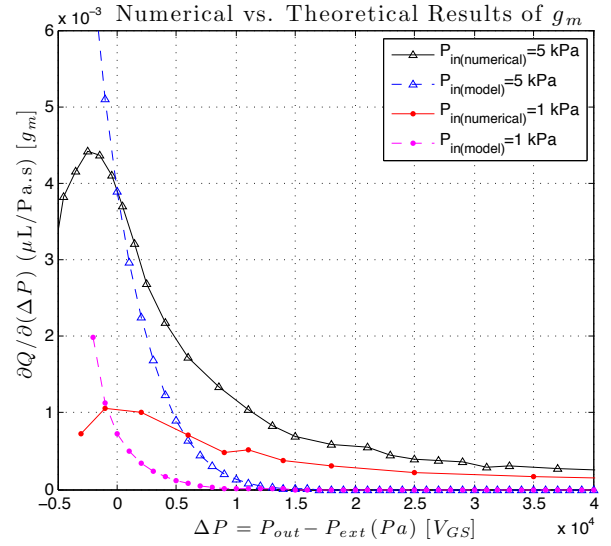


FIGURE 10. COMPARISON OF EXPERIMENTAL AND THEORETICAL RESULTS OF THE FLUIDIC TRANSCONDUCTANCE

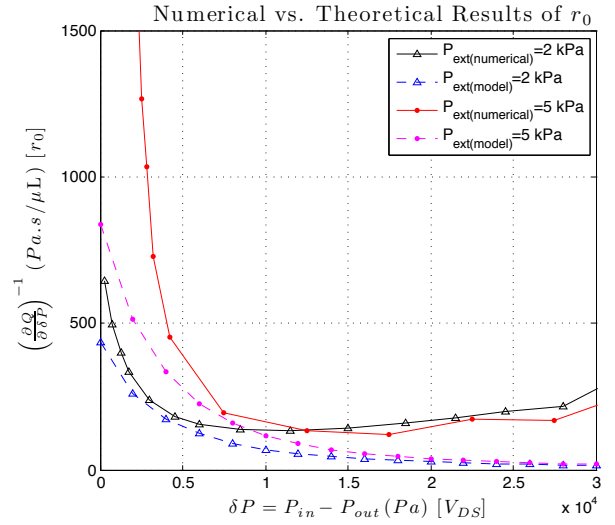


FIGURE 11. COMPARISON OF EXPERIMENTAL AND THEORETICAL RESULTS OF THE INTRINSIC RESISTANCE

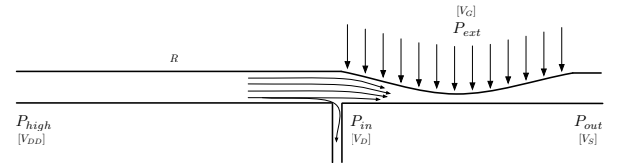


FIGURE 12. COMMON-SOURCE FLUIDIC CIRCUIT

of the transconductance and the total output resistance; which is the resistance r_o of the transistor in parallel to the resistance r_L of the channel. Accordingly, for small deflection, we get the total gain $\bar{A}_v$

$$\bar{A}_v = g_m(r_o || r_L) \approx \frac{3}{2} \frac{L_c}{h_0 C} \frac{\delta p}{L_c + \frac{3L_c}{h_0 C} \left(3 \frac{\delta p}{4} + \Delta P\right) + L \sqrt{1 + \frac{3}{h_0 C} \left(\frac{\delta p}{2} + \Delta P\right)}}$$

where L_c is the length of the channel before the transistor. g_m and r_o can be extracted from the simulation graphs presented previously or alternatively it can be obtained from Eqn. (18). The pressure at the exit of the channel (input of transistor), as determined from transient simulation of the fluid-structure interaction on ANSYS/FLUENT, is plotted versus time along with the applied external pressure in Fig. 13, for the case where $p_{out} = 100$ kPa, $p_{ext} = 100 + 5 \sin(2\pi f t)$, $f = 2$ Hz. It can be observed from Fig. 13 that p_{in} assumes a harmonic sinusoidal dependence on time, similar to p_{ext} , which indicates that the device is linear, as expected from small signal behavior. The pressure gain of the common-source configuration is $p'_{in}/p'_{ext} = 1.75/5 = 0.35$. The

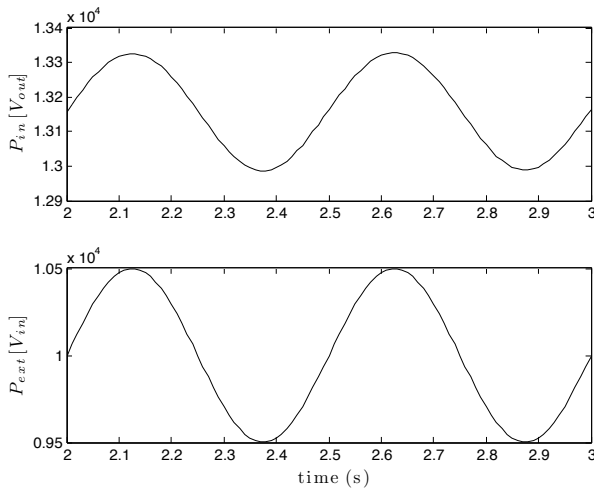


FIGURE 13. EFFECT OF p_{ext} ON p_{in} FOR LOW FREQUENCIES (2Hz)

analysis can be extended for a range of frequencies to characterize the effect of frequency on gain for low frequencies. It is therefore, noticed in Fig. 14 that the pressure gain has no dependence on the frequency which is similar to the behavior of a MOSFET for low frequencies. It remains to be seen whether this independence still holds at higher frequencies.

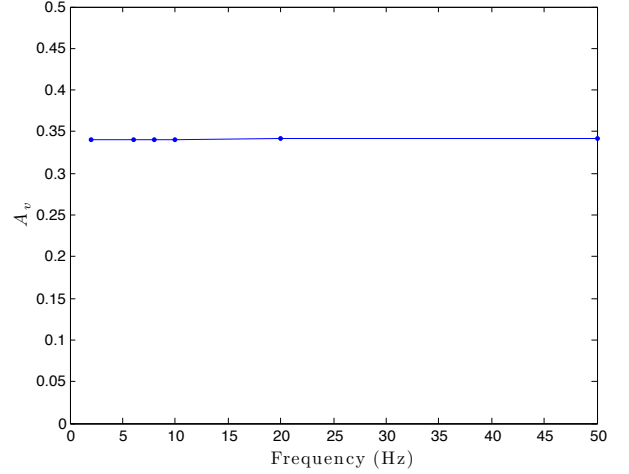


FIGURE 14. DEPENDENCE OF TRANSCONDUCTANCE ON FREQUENCY

CONCLUSION

A reduced order small signal analog model of a microfluidic transistor is presented. Characterization of the microfluidic transistor based on steady characteristic curves of the volume flowrate versus pressure drop and external pressure minus output pressure is presented in a manner analogous to $I_D - V_{GS}$ and $I_D - V_{DS}$ characteristic curves for a MOSFET. The reduced order model parameters (transconductance and intrinsic output resistance) can be derived from analytical models similar to the one presented or more accurately from numerical simulation of the coupled fluid-structure problem. Extraction of these small signal parameters allows for fast small signal type analyses. Surprisingly, the proposed microfluidic transistor is similar in many respects to its electronic analog. Both have a square dependence of the current (volume flowrate) on the gate-source voltage (external pressure minus output pressure) in the long channel operation and a linear dependence in the short channel operation. Both also have a triode and saturation regime depending on the value of the pressure difference across the channel (V_{DS}). The benefits of accurate modeling and characterization of microfluidic transistors allows them to evolve from operating as binary on/off switches to analog component that can provide a variable gain. It is expected that this analog DC and small signal characterization will play an important role in future complex microfluidic devices as it offers accurate control of both amplitude and frequency of the pressure and volume flow rate.

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